

RAPID ORGANIC CANADA

WP-003

How to Build a Defensible Organic-Waste Business Case

A measurement-first framework for baselines, pilot KPIs, total cost, ROI and investment decisions

DOCUMENT POSITION

This paper explains how to evaluate an on-site organic-waste project without treating estimates as evidence. It is a decision framework—not a quotation, performance guarantee, financial recommendation, regulatory approval or site-specific engineering analysis.

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HOW TO READ THIS PAPER

The method separates facts, measured pilot results, supplier specifications and assumptions. The business case becomes stronger when an unfavourable result is allowed.

Purpose

WP-003 provides a common decision language for waste generators, engineers, finance teams, pilot partners and investors. It applies to Rapid Organic evaluations and, with suitable adaptation, to other distributed organic-waste technologies.

1. Executive summary

An organic-waste project can appear attractive while hiding weak evidence. Hauling invoices may omit internal labour. A quoted machine capacity may not equal accepted wet feedstock. Mass reduction may mainly represent water transfer. Output value may be assumed before a lawful buyer exists. Payback may ignore downtime, financing, maintenance, laboratory testing and residual disposal.

CORE PRINCIPLE

Compare the verified annual cost and risk of the current system with the verified annual cost and risk of the proposed system, using the same quantity, composition, service level and accounting boundary.

Five requirements for a defensible case

- A signed baseline built from invoices, weights, operating records and observed labour.
- A declared system boundary that prevents costs or emissions from disappearing between scenarios.
- A pilot protocol with calibrated measurements, acceptance criteria and complete deviation records.
- A total-cost model that uses zero—not hoped-for revenue—where no enforceable output agreement exists.
- Sensitivity and downside cases that expose which assumptions control the decision.

Decision output

The result should be a short investment memorandum supported by an auditable workbook and evidence register. Its recommendation may be proceed, proceed with conditions, revise and retest, or stop. “Stop” is a valid value-creating outcome when it prevents an unsuitable purchase.

2. Why business cases fail

Failure mode	Why it distorts the result	Required correction
Invoice-only baseline	Misses labour, space, cleaning, storage, incidents and management time	Map the complete current process and reconcile records
Nameplate capacity	Confuses nominal rating with accepted material under real conditions	Report accepted kg/day, cycles, utilization and rejected loads
Unreconciled mass reduction	May count evaporated or transferred water as eliminated waste	Close a mass and moisture balance
Free output revenue	Assigns value before quality, status, logistics and demand are proven	Use zero until a buyer or avoided-cost route is documented
Perfect availability	Removes downtime, service response and contingency hauling	Use measured availability and downside cases
One utility number	Hides peak demand, ventilation, preprocessing and cleaning energy	Meter the defined system boundary
Simple payback only	Ignores asset life, timing, financing and terminal cost	Add cash flow, NPV, IRR and sensitivity

WARNING SIGN

If the proposal becomes unattractive when one optimistic assumption is removed, it is not yet an investment-grade business case.

Evidence hierarchy

- Highest: calibrated site measurements, source records, invoices and independent laboratory results.
- Strong: controlled pilot data tied to a frozen configuration and disclosed conditions.
- Limited: comparable-site data with documented differences.
- Weakest: supplier estimates, industry averages and unsupported promotional claims.

3. Define the decision and system boundary



A defensible result preserves source records, assumptions, exclusions and uncertainty.

Figure 1. Evidence flow for an investment-grade organic-waste decision.

Write the decision first

State exactly what approval is being requested: a feasibility study, a time-limited pilot, a first commercial unit, or a multi-site rollout. Each requires a different evidence threshold. Do not use pilot evidence to imply fleet-scale economics without an explicit scale model.

Boundary item	Baseline scenario	Proposed scenario
Material	Generation point to current final destination	Generation point through processing to actual output destination
Time	At least one representative operating period	Same season or a normalized equivalent
Service	Current pickups, storage and compliance level	Equivalent or improved service, including downtime contingency
Utilities	Existing refrigeration, compacting or pretreatment	Equipment, preprocessing, ventilation, air treatment, cleaning and controls
Labour	Collection, bin handling, cleaning, records and supervision	Loading, monitoring, discharge, cleaning, maintenance and records
Externalities	Incidents and measurable operational exposure	New odour, wastewater, safety, permitting and output risks

FUNCTIONAL UNIT

Use one common denominator—normally cost per accepted wet tonne, plus annual totals. Add cost per tonne of dry solids or useful output when moisture changes materially.

4. Build the baseline

The baseline is the counterfactual: what the organization will spend and experience if the project is not installed. It should be owned jointly by operations and finance, not reconstructed by the technology supplier alone.

Data group	Minimum records	Normalization
Material	Scale tickets, bin counts, production linkage, rejected loads, composition and moisture samples	kg/day; kg/unit production; seasonal profile
Logistics	Pickup frequency, kilometres if known, minimum charges, fuel surcharges and emergency service	\$/accepted tonne and \$/year
Facility	Bin rental, cold storage, floor area, drains, cleaning supplies and pest control	\$/year and hours/month
Labour	Collection, segregation, moves, cleaning, administration and supervision	loaded labour \$/tonne
Disposal / recovery	Tipping or processing fees, rebates, contamination penalties and contract escalation	\$/tonne by route
Risk / service	Missed pickups, spills, complaints, shutdowns and production disruption	frequency, cost and documented consequence

Baseline quality checks

- Reconcile total annual tonnes across generator, hauler and receiver records.
- Separate fixed charges from variable charges; both affect savings differently.
- Record taxes consistently and distinguish recoverable taxes.
- Exclude sunk cost, but include avoidable current cost and future committed cost.
- Document the representative period and explain strikes, outages, peaks or abnormal production.

BASELINE SIGN-OFF

Finance confirms cost records; operations confirms the process map and labour; environmental staff confirms current destinations and compliance conditions.

5. Design a measurement-grade pilot

Phase	Required work	Release gate
Protocol	Freeze machine, software, feedstock envelope, sampling plan, instruments and acceptance thresholds	All parties approve before data collection
Baseline confirmation	Measure representative input, handling and current route	No pilot comparison without usable baseline
Commissioning	Verify utilities, safety functions, controls, training and contingency route	No production loads until signed release
Controlled runs	Vary only planned factors; preserve raw time-series, samples and deviations	Repeatability within agreed envelope
Operational run	Normal operators and schedule; track uptime, labour, cleaning and interruptions	Sufficient duration and representative conditions
Review	Independent data check, laboratory results, economics and limitation statement	Proceed, revise/retest or stop

Essential instruments and records

- Traceable input and output scales; moisture or total-solids testing.
- Dedicated electricity metering and other utility measurement as applicable.
- Cycle log, alarms, downtime reason codes and maintenance work orders.
- Operator labour records separated into routine, cleaning and corrective work.
- Batch identification, sample chain of custody and actual final disposition.

DATA OWNERSHIP

The pilot agreement should preserve complete raw data and define review, confidentiality and publication rights before the first run.

6. Normalize technical performance

Metric	Recommended expression	Interpretation caution
Accepted throughput	accepted input kg / scheduled operating day	Do not include rejected or bypassed loads
Utilization	actual accepted input / demonstrated available capacity	Low utilization can dominate economics
Availability	run-ready time / scheduled required time	Define planned maintenance consistently
Specific energy	metered kWh / accepted wet tonne; also / dry-solids tonne	Wet-tonne results can reward dilution
Labour intensity	routine + cleaning + maintenance hours / accepted tonne	Separate learning-curve labour
Net material routed	all solid and liquid outputs by destination / input	Nothing leaves the mass balance
Output yield	characterized output kg / accepted input kg	State wet and dry basis

Mass-balance equation

CONSERVATION CHECK

Input mass = solid output + captured liquid + condensate + estimated exhaust moisture + samples + cleaning-associated transfer + unexplained difference.

Set an acceptable closure range in the protocol based on instrument uncertainty and sampling. A large unexplained difference invalidates claims about reduction, diversion and output yield.

Representative operation

Results should span the approved feedstock range, ordinary staff and realistic production variability. Report the median and range, not only the best batch. Where scale-up is proposed, identify which costs scale with throughput, which remain fixed and which require redesigned equipment.

7. Calculate total cost of ownership

Cost class	Include
Initial capital	Equipment, freight, duties, site engineering, permits, foundations, electrical, ventilation, drainage, fire/life safety, commissioning and training
Recurring operations	Electricity, water, additives, filters, consumables, cleaning, routine labour, laboratory work, insurance and administration
Maintenance	Planned service, wear parts, calibration, software/support, travel and corrective repairs
Residual logistics	Output containers, storage, hauling, receiver fees, rejected loads and off-spec material
Downtime	Contingency hauling, rental equipment, production disruption and restart labour
End of life	Decommissioning, restoration, disposal and any residual value

Annualized comparison

ANNUAL NET BENEFIT

Avoided baseline cost + contracted output value + separately supportable incentives – annual operating cost – residual logistics – annualized capital/compliance cost.

Avoided cost must truly disappear. If a pickup contract has a fixed minimum, reducing tonnes may not reduce annual cost proportionally. Similarly, internal labour is a saving only if time can be eliminated or redeployed to measurable value.

Do not double count

- Do not count avoided disposal and output revenue for the same tonne unless both occur and are documented.
- Do not count depreciation as a cash outflow while also counting the full capital purchase in cash flow.
- Do not add carbon value, grant funding or tax benefit unless eligibility and ownership are documented.
- Do not treat working-capital timing as operating profit.

8. Model value without double counting

Value category	Recognition rule	Conservative default
Avoided hauling / tipping	Use contract terms and demonstrated change in service	Only the variable portion that will disappear
Avoided storage / handling	Use a real cancelled cost or redeployed constrained resource	Zero if the resource remains and has no alternate use
Output sale	Executed offtake terms, accepted specification and net logistics	Zero
Output avoided fee	Receiving route has confirmed acceptance and price	Current contingency disposal cost
Production benefit	Measured causal improvement under comparable conditions	Zero
Grant / incentive	Written eligibility and probability-weighted timing	Zero until approved
Carbon / ESG value	Declared method, boundary, ownership and verification route	Report physically; no cash value

Output valuation ladder

- 1. Liability: the output costs money to store, test or dispose.
- 2. Neutral route: accepted at no fee, but transport and handling remain.
- 3. Avoided cost: used internally and demonstrably replaces a purchased input.
- 4. Contracted product: third party accepts a specification at a defined net price.

INVESTMENT RULE

A project that works only when unvalidated output is priced as premium fertilizer is not commercially validated.

9. Financial evaluation and sensitivity

Financial analysis converts technical and commercial evidence into dated cash flows. The discount rate, tax treatment, financing and accounting rules should be set by the host organization. Rapid Organic should provide transparent inputs, not prescribe the customer's investment policy.

Measure	Use	Limitation
Simple payback	Quick screen: initial investment / annual net cash benefit	Ignores timing after payback and discounting
Net present value (NPV)	Present value of all incremental cash flows at the host discount rate	Sensitive to forecast horizon and terminal assumptions
Internal rate of return (IRR)	Rate at which NPV equals zero	Can mislead with unusual or multiple-sign cash flows
Equivalent annual cost	Compares options with different lives on an annual basis	Depends on discount rate and service equivalence
Break-even value	Finds the threshold for energy, utilization, output cost or uptime	Does not show probability by itself

Required scenarios

Case	Purpose
Downside	Lower utilization and availability; higher energy, maintenance and output cost; no output revenue
Base	Best-supported values from representative pilot evidence
Upside	Supportable improvement—not the supplier's theoretical maximum
Break-even	Solve for the single variable that makes NPV or annual benefit equal zero

SENSITIVITY FOCUS

Rank variables by their effect on NPV and payback. If utilization, output route or uptime dominate, resolve those uncertainties before negotiating minor equipment-price changes.

10. Environmental claims and carbon boundaries

Organic material disposed in landfill can generate methane, so diversion can create environmental value. The size of that value depends on the baseline landfill, gas capture, transport, electricity, material composition, alternative treatment and output fate. A business case should not convert tonnes “diverted” directly into emissions avoided without a defined method.

Claim	Evidence needed
Waste diverted	Mass by verified destination; exclude storage and unapproved accumulation
Transport reduced	Actual baseline and alternative trips, vehicle/load data and route distance
Energy impact	Metered utilities and disclosed emission factors
Avoided landfill emissions	Recognized model with feedstock, landfill and gas-management assumptions
Circular output	Characterized output plus documented beneficial use that replaces or supports a defined function
Life-cycle improvement	Goal, scope, functional unit, inventory, allocation and interpretation consistent with the chosen LCA method

Canadian public tools

Environment and Climate Change Canada provides a greenhouse-gas calculator that compares a baseline with alternative organic-waste management scenarios. It is useful for structured screening, but project claims still require disclosed inputs, version, boundary and interpretation. Canada’s Fuel Life Cycle Assessment Model methodology identifies ISO 14040 and ISO 14044 as the relevant LCA framework and requirements.

SEPARATE LEDGERS

Keep physical environmental results, verified carbon claims and financial carbon value as three separate records. They are not interchangeable.

11. Risk, uncertainty and evidence quality

Risk	Control before approval
Feedstock variation	Released acceptance envelope, sampling and rejection route
Performance scale-up	Frozen reference design, scaling assumptions and staged capacity test
Output classification / market	Laboratory dossier, regulatory review and contingency receiver
Reliability / service	Availability evidence, spares, response time, warranty and bypass plan
Site interface	Utility survey, ventilation, drainage, fire/life safety and permitting review
Financial	Quoted installation scope, contingency, escalation, financing and downside model
Claim / reputation	Evidence register, approved wording and correction process

Evidence register

- Assign each input an owner, source, date, unit and confidence rating.
- Link every headline result to the supporting record or formula.
- Mark supplier estimate, customer record, measured pilot value and independent result distinctly.
- Version-control the model; record every assumption changed after review.
- State exclusions and unresolved uncertainty in the recommendation itself.

CONFIDENCE RULE

Precision in a spreadsheet does not improve weak evidence. Round outputs consistently with the uncertainty of the inputs.

12. Decision gates and procurement package

Gate	Minimum evidence	Possible decision
A — Problem verified	Signed baseline and defined decision	Stop or authorize feasibility
B — Fit verified	Feedstock screen, site survey, approvals path and contingency route	Stop, redesign or authorize pilot
C — Performance verified	Representative pilot, mass balance, utilities, labour, uptime and outputs	Stop, extend or build commercial model
D — Value verified	TCO, downside case, lawful output route and customer acceptance	Reject, negotiate or approve first unit
E — Replication verified	Reference operation, configuration control, service and manufacturing evidence	Hold or scale

Procurement-ready package

- Basis-of-design and site responsibility matrix.
- Released feedstock envelope and acceptance/rejection procedure.
- Guaranteed and non-guaranteed performance terms with test method.
- Installation scope, commissioning criteria and operator training.
- Warranty, exclusions, service response, critical spares and data access.
- Output responsibility, off-spec route and regulatory responsibilities.
- Commercial model with evidence register, scenarios and approval signatures.

CONTRACT ALIGNMENT

The contract should test the same metrics and boundaries used to justify the investment. A business case cannot be “guaranteed” by unrelated nameplate specifications.

13. Worked example

The following fictional example demonstrates method only. It is not a Rapid Organic quotation or performance claim.

Input	Illustrative base value	Source class
Organic residual	250 wet tonnes/year	Customer scale records
Avoidable hauling and receiving cost	\$190/tonne	Contract and invoices
Other avoidable handling cost	\$8,000/year	Time study and cancelled services
Installed project cost	\$240,000	Budget quotation
Annual operating and maintenance cost	\$24,000/year	Pilot-normalized estimate
Residual/output logistics	\$7,500/year	Receiver indication
Contracted output revenue	\$0	No contract
Evaluation life	8 years	Host policy
Discount rate	8%	Host policy

Base calculation

Avoided annual cost = $(250 \times \$190) + \$8,000 = \$55,500$. Annual net operating benefit before capital = $\$55,500 - \$24,000 - \$7,500 = \$24,000$. Simple payback = $\$240,000 / \$24,000 = 10.0$ years. Over an eight-year evaluation life, this base case does not recover the initial investment on an undiscounted basis; its NPV is therefore negative before any terminal value.

What the example teaches

- A real waste problem does not automatically create an investable project.
- Adding speculative output revenue could manufacture an attractive payback but would weaken the case.
- The next work should target installed cost, utilization, verified avoided cost and operating cost—not improve the presentation.
- A service, lease, shared-savings or different-scale model may be better than a capital sale, but each requires its own cash-flow analysis.

HONEST CONCLUSION

On these illustrative inputs, do not approve the project as presented. Revise the configuration or commercial model and retest the variables that control value.

14. Frequently asked questions

How long should the baseline period be?

Long enough to capture the operating and seasonal variation that controls quantity, composition and cost. Four to eight weeks may support a narrow pilot, while an annual investment case usually needs a full-year record or a defensible normalization.

Can a supplier estimate be used?

Yes, if it is clearly labelled, sensitivity-tested and replaced by measured evidence before final approval. It should never be presented as a verified site result.

Should internal labour count as savings?

Only when the work will actually disappear or the released time has a credible alternative value. Reduced minutes do not automatically equal cash savings.

Can output be assigned fertilizer value?

Not until its composition, safety, regulatory status, consistency, logistics and buyer acceptance support that value. The conservative pre-validation value is zero or the actual cost of lawful disposition.

Is mass reduction the same as diversion?

No. Water or other material may move to air, drains or another route. Diversion depends on verified final destinations and the declared boundary.

Is simple payback enough?

It is useful for screening, but final approval should also consider cash-flow timing, asset life, NPV, uncertainty and downside exposure.

Who should own the model?

The customer should own the investment decision and baseline. The supplier should provide traceable technical and commercial inputs. Finance, operations and environmental staff should sign their respective data.

What is Rapid Organic's best next proof point?

A representative pilot that links complete technical data to a signed customer baseline, a lawful output route and an approved before/after case study.

15. Conclusion

A defensible organic-waste business case is not a persuasive estimate. It is a traceable comparison between equivalent current and proposed systems. It begins with the customer's real baseline, measures a defined pilot, reconciles all material and utility flows, includes the full lifecycle cost, values outputs conservatively and exposes downside conditions before capital is committed.

RAPID ORGANIC POSITION

Rapid Organic's commercial credibility should be built one verified baseline and one transparent pilot at a time. The objective is not to make every site appear attractive; it is to identify the sites where measurable technical performance creates durable customer value.

Selected public references

- [Environment and Climate Change Canada — Waste and greenhouse gases: Canada's actions](#)
- [Environment and Climate Change Canada — Greenhouse gas calculator for organic waste management](#)
- [Environment and Climate Change Canada — Fuel Life Cycle Assessment Model methodology](#)
- [Rapid Organic — WP-001 Technology Overview and Value Proposition](#)
- [Rapid Organic — WP-002 On-Site Organic Waste Processing for the Seafood Industry](#)

Document control

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